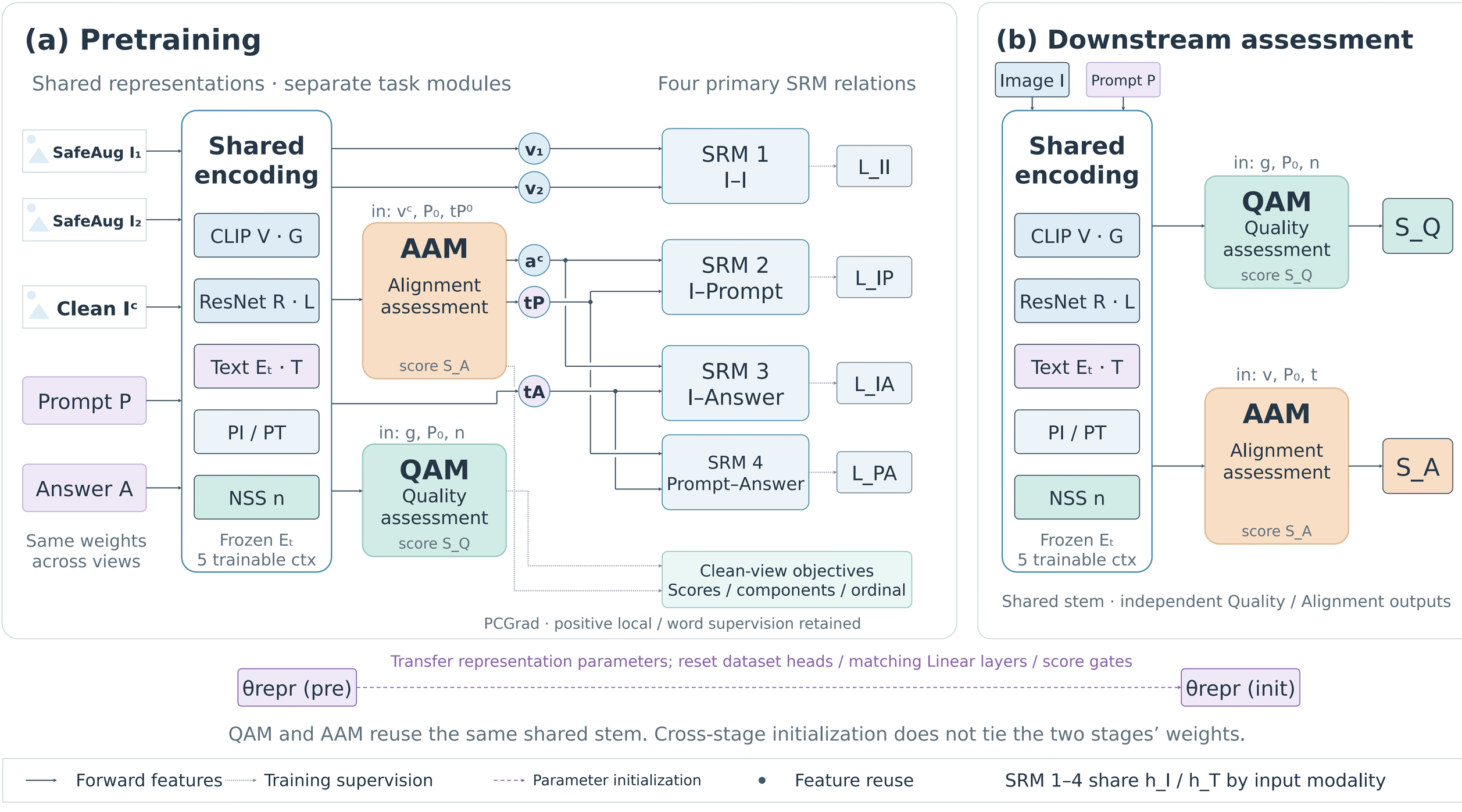
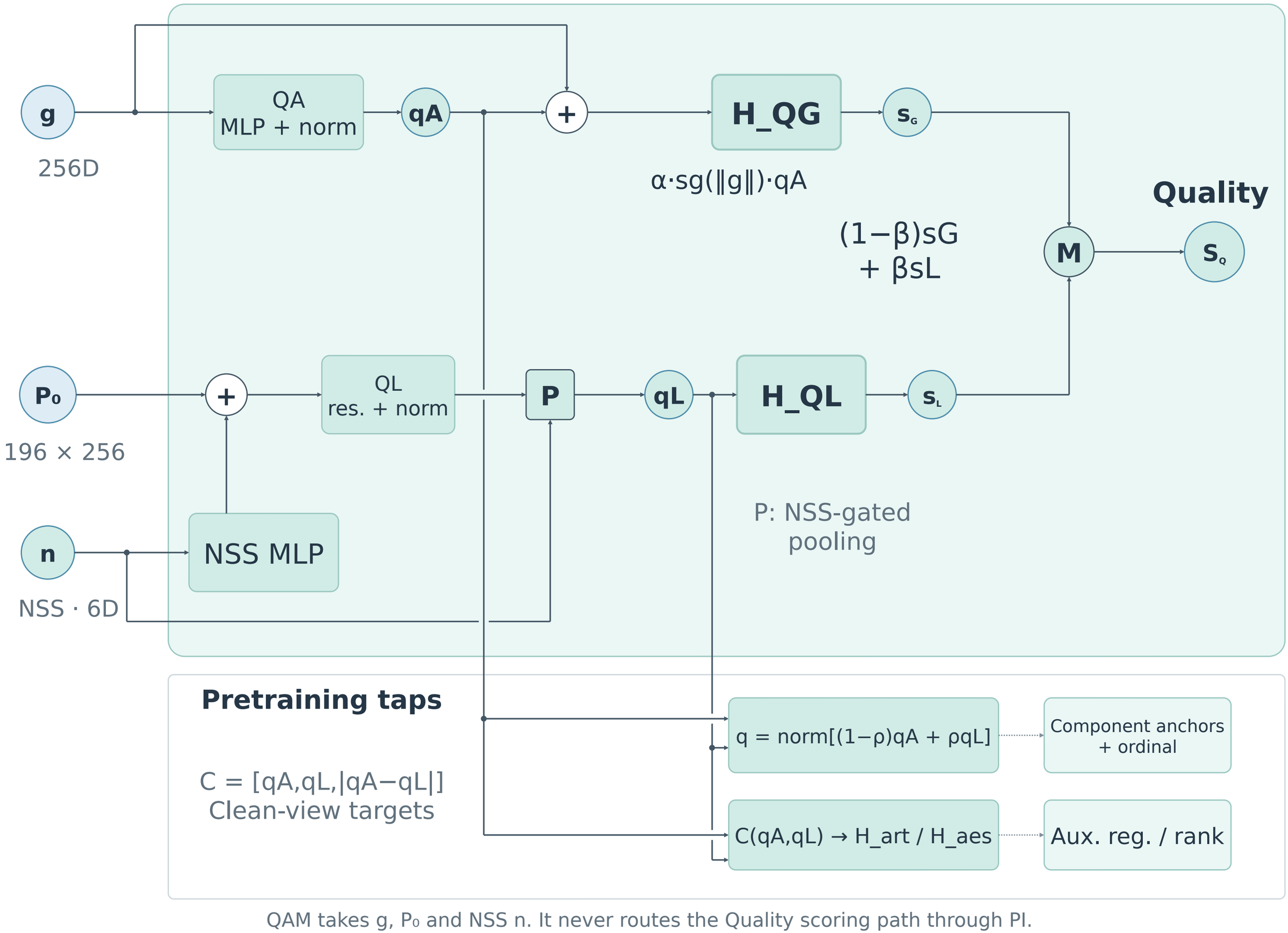
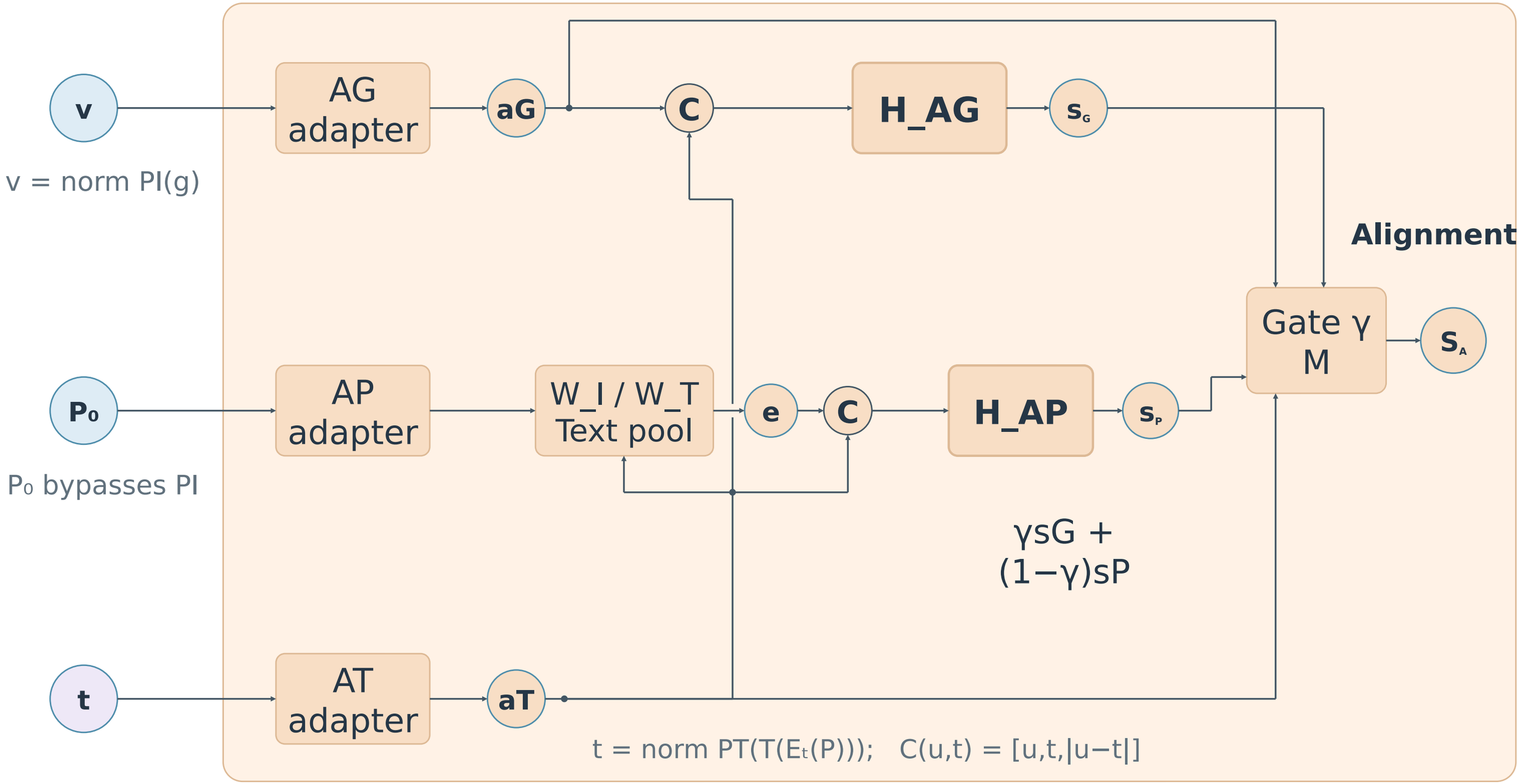

(a) Pretraining
Shared representations · separate task modules
Four primary SRM relations
SafeAug I₁
SafeAug I₂
Clean I^c
Prompt P
Answer A
Same weights across views
Shared encoding
CLIP V · G
ResNet R · L
Text Et · T
PI / PT
NSS n
Frozen Et
5 trainable ctx
v₁
v₂
in: v^c, P₀, tP⁰
AAM
Alignment assessment
score S_A
a^c
tP
tA
in: g, P₀, n
QAM
Quality assessment
score S_Q
SRM 1
I–I
SRM 2
I–Prompt
SRM 3
I–Answer
SRM 4
Prompt–Answer
L_II
L_IP
L_IA
L_PA
Clean-view objectives
Scores / components / ordinal
PCGrad · positive local / word supervision retained
(b) Downstream assessment
Image I
Prompt P
Shared encoding
CLIP V · G
ResNet R · L
Text Et · T
PI / PT
NSS n
Frozen Et
5 trainable ctx
in: g, P₀, n
QAM
Quality assessment
score S_Q
S_Q
in: v, P₀, t
AAM
Alignment assessment
score S_A
S_A
Shared stem · independent Quality / Alignment outputs
Transfer representation parameters; reset dataset heads / matching Linear layers / score gates
θrepr (pre)
θrepr (init)
QAM and AAM reuse the same shared stem. Cross-stage initialization does not tie the two stages' weights.
Forward features
Training supervision
Parameter initialization
Feature reuse
SRM 1–4 share h_I / h_T by input modality

QAM · Quality Assessment Module



QAM takes g, P₀ and NSS n. It never routes the Quality scoring path through PI.

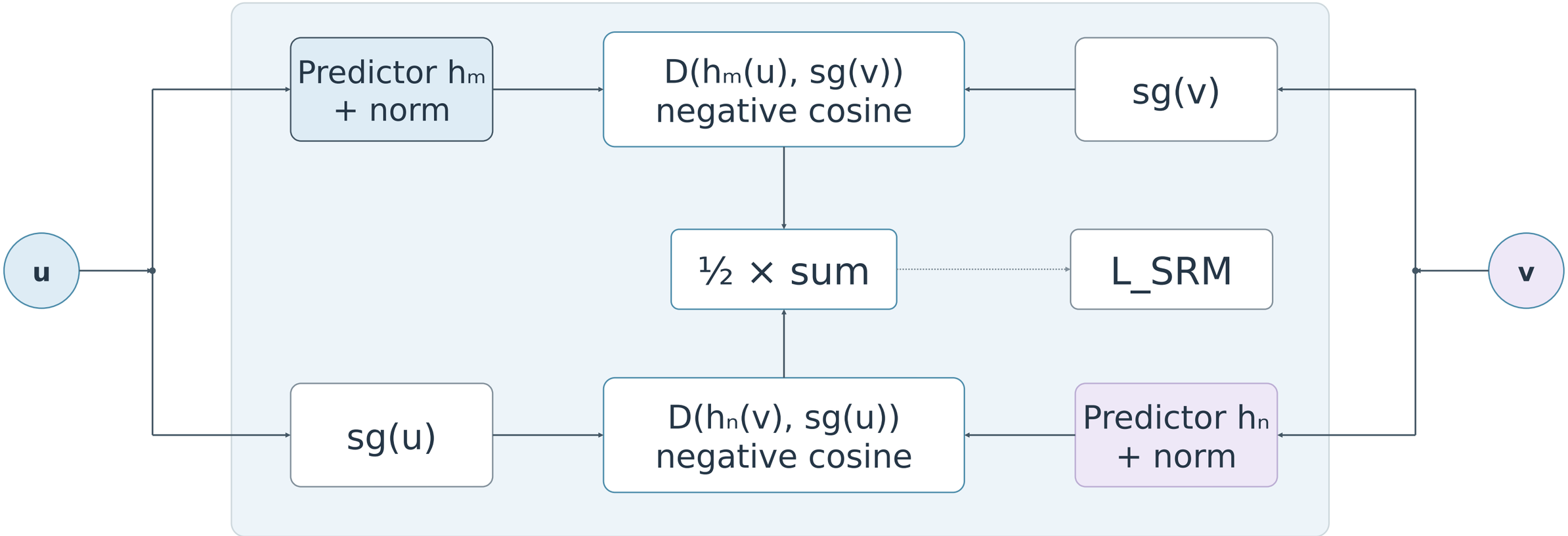
AAM · Alignment Assessment Module



Training taps: a^c = norm(AG(v^c)); tP = norm(W_T(aT)). Answer tA comes directly from PT, outside AAM.

SRM · SimSiam Relation Module

Feature inputs from the shared stem / AAM · predictors are reused by modality



Four primary relations · two shared modality-specific predictor parameter sets

SRM 1 · I–I	SRM 2 · I–Prompt	SRM 3 · I–Answer	SRM 4 · Prompt–Answer
Inputs: v_1 , v_2	Inputs: a^c , tP	Inputs: a^c , tA	Inputs: tP , tA
Predictors: h_I / h_I	Predictors: h_I / h_T	Predictors: h_I / h_T	Predictors: h_T / h_T

PI/PT remain in the shared stem and transfer downstream; h_I/h_T are pretraining-only predictors (256 → 128 → 256).